African Agricultural Market Price Monitor

Interactive dashboard analysing food commodity prices across 6 African countries using SQL & Power BI.

Project Overview

Developer: Nhambe DB

Tools: SQL (PostgreSQL) & Microsoft Power BI

Data Source: WFP Global Food Prices Dataset (Kaggle)

Sector: Agricultural Market Intelligence / Food Security

Executive Summary

This dashboard was designed to monitor and analyse agricultural commodity price trends across six African countries between 2017 and 2021. By combining SQL-based data extraction with Power BI visualisation, the tool enables policymakers, agribusinesses, and market analysts to track price volatility, identify the most expensive markets, and compare pre- and post-2020 price shifts, a period heavily influenced by the COVID-19 pandemic.

Problem Statement

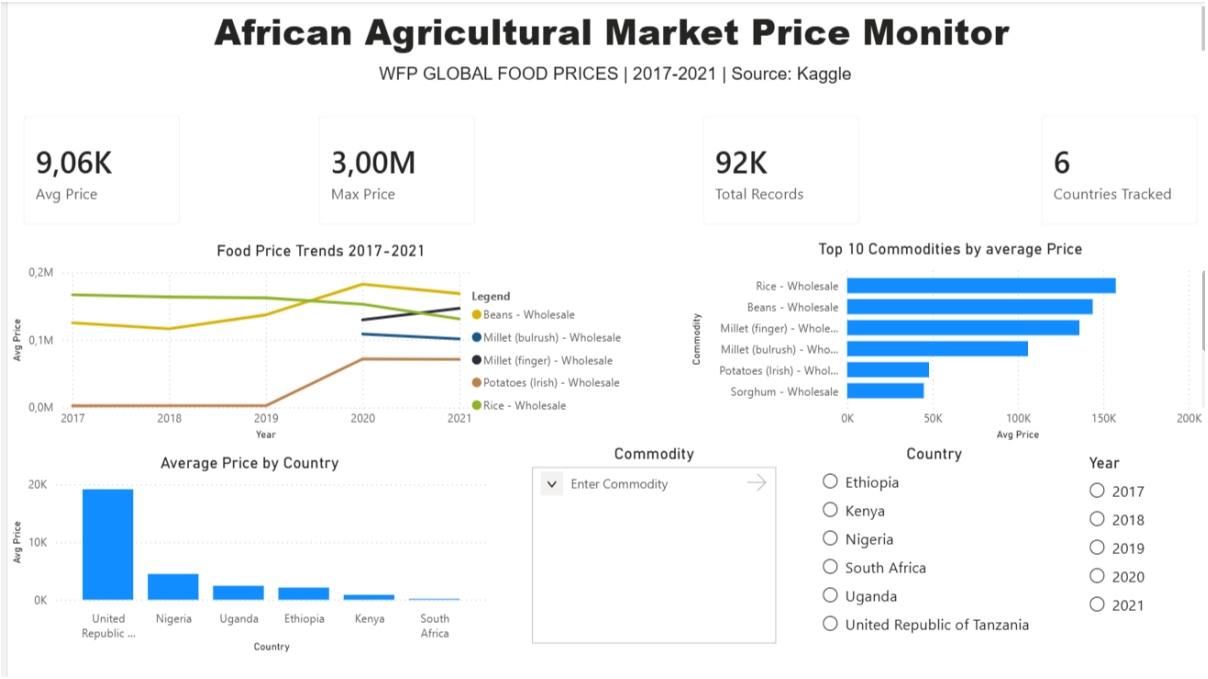
Agricultural commodity prices in Africa are highly volatile and vary significantly across countries and markets. Without a centralised monitoring tool, it is difficult for stakeholders to identify which commodities are driving food insecurity, which markets are consistently overpriced, and how prices have shifted over time. This lack of visibility makes planning, procurement, and policy decisions reactive rather than proactive.

Solution & Technical Implementation

- **SQL Data Extraction:** Five structured queries were written to extract meaningful insights from the raw WFP dataset, including average price per commodity, most expensive market per country, maize price trends over time, top 10 countries by average food price, and a pre- vs post-2020 price comparison.
- **Power BI Dashboard:** Findings were visualised in an interactive single-page dashboard featuring KPI cards (Avg Price, Max Price, Total Records, Countries Tracked), a food price trend line chart (2017-2021), a top 10 commodities bar chart, and an average price by country bar chart.

- **Interactive Filters:** Users can slice the data by commodity, country, and year, enabling targeted market analysis without any technical knowledge.
- **Scoped Dataset:** The global WFP dataset was filtered down to six African countries, Ethiopia, Kenya, Nigeria, South Africa, Uganda, and United Republic of Tanzania, to keep the analysis contextually relevant to the continent.

Dashboard screenshot



Key Impact

- **Price Transparency:** Reveals which commodities (Rice, Beans, Millet) consistently command the highest prices across African markets.
- **Market Risk Identification:** Highlights the most expensive markets per country, helping procurement teams avoid high-cost supply sources.
- **COVID-19 Price Shift Analysis:** The pre- vs post-2020 comparison quantifies how the pandemic disrupted food pricing across the continent.
- **Food Security Insight:** Equips NGOs, governments, and agribusinesses with the data needed to respond to food price crises proactively.